Week 22 Homework

Build a Literature-Based Baseline for Your Week 21 Project

Use prior work to make your project question clearer, fairer, and more feasible.

Student: _____ **Date:** _____

Your task

Continue the **same community project and dataset proposal you developed in Week 21**. Find and read **three relevant research papers, technical reports, or authoritative project sources**. Complete the source table, summarize what the sources establish, and use them to define a reasonable comparison baseline for your own research plan.

Week 21 project idea → relevant prior work → evidence gap → baseline and next question

Estimated effort: **90–120 minutes**. Submit the completed table and short synthesis at the next class, unless your mentor sets another deadline. **No code, finished model, or new experiment is required.**

What counts as a source?

Use a source that directly informs your question, data, method, or community context. Suitable sources include peer-reviewed research papers, government or utility technical reports, and documented public research projects. For each source, open and inspect the actual paper or report; a search-result snippet, AI summary, or homepage alone is not sufficient. Aim for sources you can understand and verify. If you cannot access the full source, record the problem and select another.

Source-quality check: Who produced it? What question does it address? What data or evidence does it use? Does it explain its method and limits? Is the result relevant to your place, people, time period, and proposed decision?

Part A: Set your project context

Week 21 project title:

Your bounded research question: Name a place or group and a time period where possible.

Intended user and possible decision: Who might use your result, and what could they decide?

Dataset you proposed: Give its title and publisher. What do its rows represent?

Part B: Literature and evidence table

Complete one row for each source. Use short, accurate notes rather than copying an abstract. Include units and the comparison method where the source gives them. If a field is not reported, write "not reported" rather than guessing.

Source and year	Question and context	Data and method	Comparison / baseline and finding	Limitation and relevance to my project
1.				
2.				
3.				

Full citations and access details

For each source, record authors or organization, title, publication year, publication or publisher, DOI or exact source link, and the date you accessed it. Follow one citation style consistently.

Part C: What does the literature tell you?

1. Shared knowledge
What do at least two sources agree on? Point to the specific source findings; do not make a broader claim than they support.

2. Differences or uncertainty
Did sources use different locations, groups, time periods, data, methods, or baselines? How might those differences affect their results?

3. A bounded remaining question
What is still uncertain or not addressed that your Week 21 project could investigate with its proposed data? If the sources already answer your question, revise the question to make the scope or comparison clearer.

Part D: Define a baseline for your plan

The **literature baseline** describes what relevant prior work has found. Your project's **comparison baseline** is the simple or current-practice reference against which you will judge your analysis. They are related, but they are not the same thing.

- **Outcome and unit:** What measurable quantity will your project describe or predict? Examples: kWh, dollars, minutes, or prediction error.

- **Comparison baseline:** What simple rule, current practice, historical average, group median, or other fair reference will you compare with? Why is it appropriate?

- **Evidence:** Which dataset fields or observations would you need, and what time period or groups would be compared?

- **Success check:** What result would count as an improvement, and how could you verify it without claiming a benefit you have not measured?

Part E: Explain your choice and its opportunity cost

Name one realistic alternative project you considered. Explain why your Week 21 topic is the better fit for your priorities and constraints (for example, data access, time, safety, or relevance to a specific user). What is the value of the best alternative you are giving up? The alternative need not be a bad idea.

Research integrity and AI-use statement

Use AI only as a limited assistant for search-term ideas, plain-language clarification, or feedback on your own notes. Open and verify every source yourself. Do not submit invented citations, quotations, methods, or results. Disclose any AI use and what you checked independently.
AI-use statement: Write "No AI used," or identify the tool, its limited role, and what you personally verified.

Final check: I used three sources I directly inspected; I separated reported evidence from my own plan; I identified a simple comparison and its limits; and I continued my Week 21 project rather than claiming results I have not measured.